WASEL · Team Briefing

ER Command Dashboard · Nexus 26, Track 3 (Informatics) · Answers to the team's 10 questions

24 October 2026
All data shown is fictional / simulated

LIVE DASHBOARD	SOURCE CODE	ARCHITECTURE DIAGRAM
wasel-er-dashboard.pages.dev	github.com/Polocobear/wasel-er-dashboard	...pages.dev/architecture

1 Framework used

- **No front-end framework.** Plain HTML, CSS and JavaScript (ES2020) with zero dependencies. It runs offline in any browser, there is nothing to install, and the code is easy to audit. A small Python build script stitches the source modules into one self-contained HTML file.
- **Face ID service:** Python with FastAPI and DeepFace (Facenet512 model, RetinaFace face detector). Enrolled faces are stored in SQLite.
- **Standards it is built on:** HL7 FHIR R4 (data model), HL7 v2.5 (messaging), IEEE 11073 (medical devices), LOINC (vital signs), ICD-10-CM (diagnoses), ESI v4 and START (triage), NEWS2 (early warning).

2 Dataset

- **No real patient data.** Every patient and name is fictional and generated by the simulation.
- **The clinical model is real:** 23 conditions with real ICD-10-CM codes, typical presenting vital signs, the published NEWS2 scoring bands (Royal College of Physicians, 2017) and real LOINC codes.
- **No training dataset is needed.** Decision support is rule-based, not machine learning.
- **The only model is face ID,** which uses DeepFace's pre-trained network. It was tested on DeepFace's own labelled test photos.
- **In a real hospital,** data would come from the EHR and the health exchanges (Riayati, Malaffi, NABIDH) over FHIR, read-only in phase 1.

3 Platforms used to build

PLATFORM	USED FOR
Claude Code	AI-assisted coding (disclose this if asked)
GitHub	Version control: github.com/Polocobear/wasel-er-dashboard
Cloudflare Pages	Hosting: wasel-er-dashboard.pages.dev (auto-deploys on every push)
Python 3	Build script, local demo server, face-ID service
JavaScriptCore	Automated tests (300+ checks)
Google Chrome	Browser testing

4 How the real-time data works

In the demo

- A simulation engine updates every half second: ambulances move, vital signs drift, beds are cleaned, patients arrive.
- The screens talk through browser messaging channels:
  - dispatch → dashboard: incidents
  - responder → dashboard: pre-alerts
  - dashboard → other screens: capacity, plus a state snapshot with names removed, every 2 s
  - other screens → dashboard: commands

In production

- Each of the 10 hospital systems pushes events into a **FHIR R4 integration gateway**:
  - HL7 v2 messages from registration
  - IEEE 11073 data from monitors and wearables
  - location tags on equipment
  - the ambulance dispatch feed
- The gateway feeds the event engine, and screens update live.

> The **Live data feed** panel on the dashboard shows every message with its standard tag, e.g. *IEEE 11073 · Observation LOINC 8867-4 Heart rate = 126*.

5 What MCI (Mass Casualty Incident) mode does

- **Triggered** automatically when an incident has 10 or more expected casualties, or declared by hand by the charge nurse or commander.
- **What changes:**
  - The mode chip turns to **MCI**.
  - Field patients get **START colours** (red / yellow / green / black).
  - A **silent surge** card appears: recall nurses, open recovery-room overflow beds, put the blood bank on standby. One confirmation does all three.
  - A **diversion** card sends walking ESI 4–5 patients to nearby EDs.
  - The **commander view** opens: incoming red patients with a readiness score, capacity gauges, staff call-back, task ownership, family reunification.
- **Stepping down:** the mode only steps down after the lower load has held for 10 minutes, so it doesn't flicker.

Steady → Strain → Surge → MCI Four modes, each triggered by readable rules

6 Sample code: real-time updates and reacting to sudden influxes

All files are in `github.com/Polocobear/wasel-er-dashboard/tree/main/src`

FILE	WHAT IT SHOWS
`10-engine.js`	The simulation: `advance()` is the real-time loop, `SCEN.pileup` is the sudden influx (22 casualties across 9 ambulances), and `updateVitals()` makes patients deteriorate
`20-ai.js`	The reactions: `aiPlan()` produces re-triage, bed placement, pre-arrival prep, supply alerts and capacity diversion. The file name is historical; the logic is rule-based
`15-wasel.js`	Operating modes, `receivePreAlert()`, `receiveIncident()`, and the allergy hard stop in `checkOrder()`
`16-readiness.js`	Trauma activation rules and the pre-arrival readiness score
`test/*.js`	300+ automated checks across 7 scenarios

Why it's realistic for a hospital IT stack

- The engine and decision logic never touch the screen, so they can run server-side and are tested headless.
- All data is shaped as FHIR resources.
- Every displayed value is escaped against code injection.
- Commands between screens are whitelisted.
- There are no third-party dependencies.

7 Cybersecurity checks

Built and working in the demo

- **Role-based access with least privilege:** 5 roles (Wall display, Nurse, Charge nurse, Physician, IT/Admin)
- **Wall display de-identified:** initials only, nothing clickable
- **Break-glass:** opening a chart outside your care needs a reason and is logged
- **Audit log** of every record view, action, role switch and command
- **Auto-lock** after 5 minutes idle
- **Escaping:** all displayed values are escaped against cross-site scripting
- **Whitelisted commands** between screens; rejected ones are logged
- **No patient data stored** in URLs or browser storage
- **Content Security Policy** on the extra pages
- **Biometric safeguards:** candidate-only, no images sent or stored, consent required to enrol
- **Kill switch** to downtime mode, plus anomaly alerts

Production design (described, not built)

- Badge-tap multi-factor login
- TLS encryption in transit and at rest
- Hosting inside the UAE
- Signed messages with replay protection
- Network segmentation
- Immutable backups

> Be honest if asked: **no external penetration test** has been done.

8 Dashboard criteria: where each rubric item is met

CRITERION	WHERE IT IS MET
UI & data integration (40%)	One clutter-free screen plus the Focus view; clinical terms (ESI, NEWS2, ICD-10, LOINC); live data; the **Data map** button and architecture page for data mapping
Code (30%)	Live simulation with 6+ scenarios, reactions to sudden influx, the security controls above, 300+ automated tests
Standards (15%)	**Standards & security** button on the dashboard, plus COMPLIANCE.md in the repo
Innovation (15%)	Pre-alerts and slider triage (communication breakdowns), nurse workload and fatigue balancing (nurse fatigue), deterioration watch and forecasting (delayed decisions)

9 Where NFC and medical history data show on the dashboard

Medical history

Click any patient row to open the **patient drawer**. From top to bottom:

- **Allergy banner,** with its source (Riayati / Malaffi / NABIDH) and date, or **UNKNOWN** if there is no record. It never says "no allergies" just because the record is empty.
- Identity status, then the field report.
- **"History · shared record":** conditions and medications (including blood thinners), each with source and date.
- The auto-written SBAR handover, then the FHIR bundle.

Before the patient arrives, safety chips appear on the pre-alert card, e.g. ⚠ *Penicillin*, *Allergies UNKNOWN*, *Anticoagulant*.

NFC wristband

- The wristband stores **only the encounter token**, never medical data.
- On arrival, the tap shows in the **Live data feed**, e.g. *NFC · Wristband T-104 tapped at ambulance bay · encounter token matched*.
- The tap links the patient to their reserved bed and record.

10 How the structure works

Four layers

10 source systems → FHIR R4 integration gateway → Event engine + rule-based decision support → Role-based screens

The 10 source systems are: EHR, registration, beds, lab, radiology, pharmacy, asset tags, monitors, ambulance dispatch and staff rostering.

Patient flow

998 dispatch → Paramedic responder (face ID + sliders) → Pre-alert → ER dashboard → Commander · kiosk · supply · equipment · admin

The decision layer suggests. A human approves each action.

Code layout

MODULE	ROLE
`00-core`	Reference data: conditions, codes, roles
`10-engine`	Simulation
`15-wasel`, `16-readiness`	Modes, pre-alerts, safety checks, trauma rules
`20-ai`	Decision rules
`30-ui`	Dashboard screens
`40-bridge`	Links the other screens to the dashboard

Diagram: **wasel-er-dashboard.pages.dev/architecture**